



kronxNBC: A Student-t Naive Bayes Classifier for Heavy-Tailed Financial Time-Series Regime Detection using R*

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Abstract:

*Financial regime classification presents substantial statistical challenges because market returns exhibit excess kurtosis, volatility clustering, and structural non-stationarity during crisis periods. Standard Gaussian Naive Bayes machine learning classifiers often fail under such conditions because the Gaussian likelihood severely underestimates the probability mass of extreme observations. This paper introduces **kronxNBC**, an R package implementing a robust Student-t Naive Bayes classifier designed explicitly for heavy-tailed financial market regime analysis within the Clock of Regimes (COR) framework. The package replaces Gaussian class-conditional densities with scaled Student-t likelihoods and estimates the degrees-of-freedom parameter through a profile likelihood grid search. This approach stabilizes estimation, prevents numerical underflow during crisis classification, and improves robustness under extreme market conditions. The package provides a complete S3 interface including model fitting, prediction, visualization, coefficient extraction, and summary methods. We present the statistical methodology, implementation details, package architecture, and empirical examples demonstrating the advantages of Student-t classification relative to Gaussian alternatives.*

Keywords: Naive Bayes, Student-t distribution, heavy tails, regime switching, financial econometrics, machine learning, robust classification, Clock of Regimes Model, R.

1 Introduction

Financial markets operate across multiple latent structural regimes characterized by fundamentally different statistical behavior. During calm periods, return distributions are approximately Gaussian and volatility remains relatively stable. During crisis regimes, however, distributions exhibit substantial excess kurtosis, asymmetric tail risk, volatility explosions, and structural instability.

Traditional Gaussian classification methods are often inadequate under these conditions because Gaussian likelihoods decay exponentially in the tails. Consequently, extreme observations receive nearly zero probability mass, producing catastrophic misclassification precisely when accurate classification is most critical.

The **kronxNBC** package addresses this problem by replacing Gaussian class-conditional likelihoods with scaled Student-t distributions. The Student-t density introduces a tail-thickness parameter ν that allows the classifier to adapt naturally to heavy-tailed crisis behavior.

The package was designed specifically for empirical financial regime detection within the Clock of Regimes (COR) framework. COR models interpret market dynamics as transitions between latent structural states with differing persistence, volatility, fragility, and survival geometry.

The principal contributions of **kronxNBC** are:

- A heavy-tailed Student-t Naive Bayes classifier for financial regime detection.
- Stable estimation of the Student-t degrees-of-freedom parameter using a profile likelihood grid search.
- Robust classification of crisis observations without Gaussian tail collapse.
- A complete S3 interface fully integrated into the R ecosystem.
- A lightweight and dependency-minimal implementation suitable for research and production workflows.

2 Statistical Background

2.1 Naive Bayes Classification

Let:

$$\mathbf{x} = (x_1, x_2, \dots, x_p)$$

represent a feature vector and let $Y \in \{1, 2, \dots, K\}$ denote the latent class label.

The Naive Bayes classifier assumes conditional independence of predictors given the class:

$$f(\mathbf{x}|Y = k) = \prod_{j=1}^p f(x_j|Y = k).$$

Applying Bayes' theorem yields:

$$P(Y = k|\mathbf{x}) = \frac{P(Y = k)f(\mathbf{x}|Y = k)}{\sum_{\ell=1}^K P(Y = \ell)f(\mathbf{x}|Y = \ell)}.$$

The Maximum A Posteriori (MAP) decision rule becomes:

$$\hat{k} = \arg \max_k \left[\log \pi_k + \sum_{j=1}^p \log f(x_j|Y = k) \right].$$

2.2 Failure of Gaussian Likelihoods

The Gaussian density is:

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

For extreme observations, the quadratic exponent drives the likelihood toward numerical zero. During crisis periods, multiple features simultaneously exhibit extreme deviations, causing the joint likelihood to collapse.

This phenomenon is particularly problematic for financial returns because empirical return distributions exhibit:

- Excess kurtosis,
- Power-law tail behavior,
- Volatility clustering,
- Regime-dependent heteroskedasticity,
- Structural non-stationarity.

2.3 Student-t Likelihood

The Student-t density is given by:

$$f_t(x|\mu, \sigma, \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\nu\pi}\sigma} \left(1 + \frac{1}{\nu} \left(\frac{x-\mu}{\sigma}\right)^2\right)^{-\frac{\nu+1}{2}}.$$

Unlike the Gaussian distribution, the Student- t density decays polynomially rather than exponentially. Consequently, extreme observations retain meaningful probability mass.

As:

$$\nu \rightarrow \infty,$$

the Student- t converges to the Gaussian distribution. Small values of ν correspond to extremely heavy tails.

3 Profile Likelihood Grid Search

Direct optimization of the Student- t degrees-of-freedom parameter is numerically unstable in many practical settings. The **kronxNBC** R package therefore employs a profile likelihood grid search over:

$$\nu \in \{3, 4, \dots, 30, 40, 60, 100\}.$$

For each candidate ν :

1. Robust location and scale estimates are computed.
2. The Student- t log-likelihood is evaluated.
3. The maximizing candidate is selected.

This approach provides several advantages:

- Numerical stability,
- Deterministic estimation,
- Elimination of optimizer convergence failures,
- Natural regularization,
- Computational transparency.

3.1 Background and Literature Review

The Naive Bayes (NB) algorithm has long been a cornerstone of machine learning, valued for its efficiency and effectiveness in classification tasks. Early contributions by ? introduced methodologies for estimating continuous distributions within Bayesian classifiers, setting the stage for adapting the Naive Bayes framework to non-Gaussian assumptions. ? analyzed the optimality of Naive Bayes under certain conditions, demonstrating its robustness even when its assumptions are not strictly met.

Stable distributions, characterized by their ability to model heavy tails and skewness, are highly relevant for datasets with non-Gaussian characteristics. ? and ? explored their applications in modeling heavy-tailed data, particularly in financial domains. ? provided computational tools for stable distributions, including the **stable** package in R, which was integral to the implementation of the Stable Naive Bayes (SNB) model (?). This package allowed for efficient parameter estimation and density computation, enabling the seamless integration of stable distributions into the Naive Bayes framework.

Lastly, ? highlighted the advantages of Bayesian methods for handling heavy-tailed data, reinforcing the potential of extending Naive Bayes to incorporate more flexible distributional assumptions. These foundational works formed the basis for our exploration of the Student-*t* distribution as an alternative to the Gaussian assumption, providing insights into enhancing the robustness and applicability of Naive Bayes-based classifiers.

4 Package Architecture

The **kronxNBC** R package follows standard R package conventions.

4.1 Core Components

The package contains the following major components:

Component	Purpose
<code>student_t_naive_bayes()</code>	Model fitting and parameter estimation
<code>predict()</code>	Posterior classification and prediction
<code>summary()</code>	Model diagnostics and summaries
<code>plot()</code>	Visualization utilities
<code>coef()</code>	Parameter extraction
<code>tables()</code>	Structured parameter tables

The source package contains:

- R source files,
- documentation generated with `roxygen2`,
- vignettes,
- empirical examples,
- unit tests,
- reproducible scripts.

5 Implementation Details

The implementation uses an Iteratively Reweighted Least Squares (IRLS)-style update for estimating location parameters. The core fitting loop evaluates:

$$w_i = \frac{v + 1}{v + z_i^2},$$

where:

$$z_i = \frac{x_i - \mu}{\sigma}.$$

Weighted updates are then computed for each candidate v . The final model object stores:

- class priors,
- location parameters,
- scale parameters,
- degrees-of-freedom estimates,
- training metadata.

6 Usage Example

6.1 Installation

Listing 1: Installing the `kronxNBC` package from a local source tarball.

```
1 install.packages("devtools")
2 devtools::install_local("kronxNBC_0.1.0.tar.gz")
```

6.2 Basic Classification Example

Listing 2: Simulating two-regime financial returns using kronxNBC.

```
1 library(kronxNBC)
2
3 set.seed(42)
4
5 n <- 300
6
7 regime <- factor(rep(c("Calm", "Stress"), each = n/2))
8
9 x1 <- c(
10   rnorm(n/2, 0, 1),
11   rt(n/2, df = 3) * 4
12 )
13
14 x2 <- c(
15   rnorm(n/2, 0, 1),
16   rt(n/2, df = 4) * 5
17 )
18
19 X <- data.frame(x1 = x1, x2 = x2)
20
21 fit <- student_t_naive_bayes(X, regime)
22
23 summary(fit)
24
25 pred <- predict(fit, X)
26
27 table(True = regime, Predicted = pred)
```

7 Empirical Financial Regime Classification

The package was designed primarily for financial market regime analysis. A typical workflow involves:

1. Computing market features:

- returns,
- realized volatility,
- drawdowns,
- skewness,
- kurtosis.

2. Constructing labeled historical regimes.
3. Fitting the Student- t Naive Bayes classifier.
4. Classifying new market states in real time.

The Student- t likelihood is particularly valuable during crisis periods because it prevents crisis observations from receiving near-zero likelihood under the Stress regime.

8 Relation to the Clock of Regimes Framework

The Clock of Regimes (COR) framework interprets financial markets as open dynamical systems transitioning between latent structural states. Within COR:

- Calm regimes exhibit low hazard and near-Gaussian behavior.
- Transitional regimes exhibit directional persistence.
- Stress regimes exhibit heavy tails, elevated fragility, and systemic instability.

The Student- t Naive Bayes classifier acts as a probabilistic regime-identification layer within this broader framework. Heavy-tailed classification is essential because stress regimes are fundamentally non-Gaussian. The **kronxNBC** R package therefore complements Hidden Markov Models, semi-Markov models, and survival-based COR operator methods.

9 Comparison with Gaussian Naive Bayes

Feature	Gaussian NBC	kronxNBC
Tail Behavior	Exponential decay	Polynomial decay
Crisis Robustness	Poor	Strong
Extreme Observations	Underweighted	Preserved
Numerical Stability	Can underflow	Stabilized
Heavy-Tail Modeling	No	Yes
Financial Crisis Detection	Weak	Strong

10 Computational Complexity

Suppose:

- n observations,
- p features,
- K classes,
- G grid points for v .

The fitting complexity is approximately:

$$\mathcal{O}(npKG).$$

Because the grid is discrete and relatively small, computation remains efficient for practical financial applications.

11 Testing and Validation

The package includes:

- unit tests using `testthat`,
- reproducible vignettes,
- empirical demonstration scripts,
- S3 method validation,
- prediction consistency checks.

Validation experiments indicate that Student- t classification remains stable even when crisis observations exceed multiple standard deviations from the class center.

12 Limitations

Several limitations remain:

- The Naive Bayes conditional independence assumption may not hold exactly.

- Regime labels often require external construction.
- The current implementation assumes static parameters.
- Online learning extensions are not yet implemented.

Future versions may incorporate:

- adaptive online estimation,
- time-varying priors,
- hidden Markov integration,
- semi-Markov regime persistence,
- multivariate Student- t dependence structures.

13 Conclusion

The **kronxNBC** R package provides a robust Student- t Naive Bayes classifier designed specifically for heavy-tailed financial market regime classification. By replacing Gaussian likelihoods with Student- t densities and employing a profile likelihood grid search for degrees-of-freedom estimation, the package addresses one of the central weaknesses of traditional Gaussian classification methods under crisis conditions.

The package integrates naturally into the R ecosystem while remaining computationally lightweight, interpretable, and statistically robust. The methodology is especially relevant for:

- financial crisis detection,
- regime-switching analysis,
- algorithmic trading,
- risk management,
- systemic fragility monitoring,
- Clock of Regimes modeling.

Availability

The source package includes:

- full source code,
- vignettes,
- empirical examples,
- reproducible scripts,
- documentation.

Acknowledgments

The R-based estimation engines were developed through an iterative human-in-the-loop process by the authors, utilizing Claude 4.6 Sonnet (Anthropic) for code debugging and documentation. All AI-generated outputs, including the implementation of the Student- t Maximization step and recursive scaling logic, were manually audited, verified for mathematical accuracy using Wolfram Mathematica 14.3, and refined by the authors to ensure alignment with the theoretical model.

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Dedication

To all people, known or unknown, who have fearlessly stood up for their beliefs for the benefit of humanity.

Masthead Graphic

The mast head collage features the historic architecture and local atmosphere of Rīga, Latvia. The main image on the left presents an Old Town vignette, capturing the historic charm and architectural character of the city. Lush, green leafy tree branches occupy the upper and left-midground, partially framing the scene and casting a natural, soft texture over the architecture. Rising prominently in the center background is the slender, towering brick spire of the historic St. Peter's Church—characteristic of Rīga's medieval skyline. Below the spire, the pale, sunlit facade of an old building features classic arched windows. A historic, dark metal street lamp post is visible in the lower left corner against a warm, terracotta-colored building base. The scene is illuminated by soft, natural daylight under a pale, slightly hazy blue sky, indicating a clear day. The top right image depicts a lively outdoor café atmosphere, where people gather in an open-air setting that reflects the social and cultural vibrancy of the surrounding urban environment. The focal point is a rustic, dark-weathered wooden barrel serving as a decorative planter. Growing out of the top are deep purple and reddish flowers. Faintly stenciled in a white, stylized script font across the front of the barrel are the words "Rīga Avenue". In the blurred background, an outdoor café terrace is visible with empty chairs, tables, and umbrellas set along a historic cobblestone street. The middle right image highlights an Art Nouveau sculptural detail, showcasing the ornate craftsmanship, flowing forms, and decorative elegance characteristic of the architectural style. A prime example of Rīga's world-famous Jugendstil (Art Nouveau) design. The symmetrical stone relief carving is of a majestic, stoic female visage. She features detailed, wavy hair parted down the center, adorned with a small crown as an ornamental headpiece. Flanking her face on both sides are large, beautifully detailed, upward-sweeping feathered wings. The sculpture is integrated directly into the limestone masonry of the building's facade, positioned just beneath a decorative horizontal ledge. The bottom right image presents a narrow street view with a window display, capturing the intimate scale, pedestrian character, and visual charm of Rīga's historic urban landscape. It features warm wooden frames, reflective glass, and historical artifacts rendered in deep red and metallic tones, contrasting with the stone masonry surrounding it.

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